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Prelab 8.1 Introduction to Machine Learning

Learning Goals

1. Identify the characteristics of an overfitted, underfitted, and well-fitted model
2. Develop, train, and refine a model that uses machine learning to detect anomalies on data batches.
3. Assess the accuracy, precision, and loss of a model in detecting anomalies on data batches.

Foundation and Background

What is Machine Learning (ML)?

Machine learning figures things out on its own. It doesn't need explicit programming to find patterns, features, or irregularities in big data sets. Compare the traditional programming paradigm to the machine learning paradigm below.

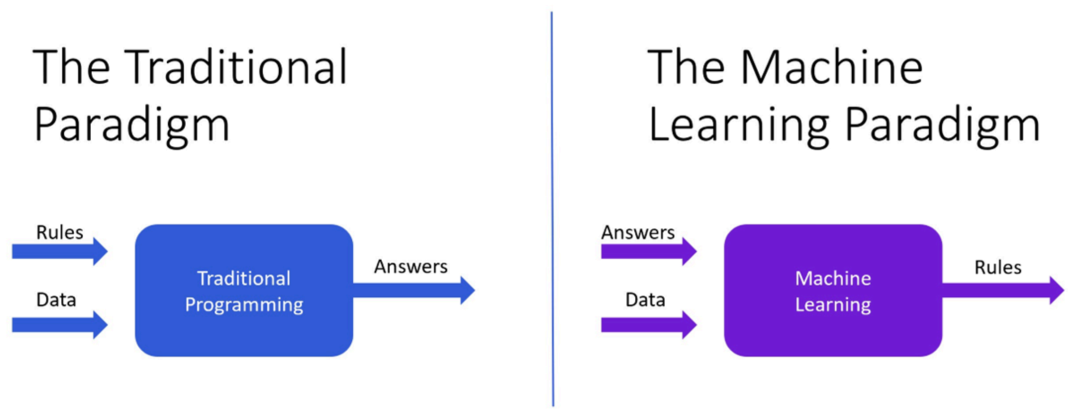


Figure 1 TP vs ML paradigm[1]

For a Machine Learning algorithm to solve a problem, there are two key features. (1) it will take a piece of **data** and (2) will take a **guess** at the **answer**. Then, it will *optimize* its **guess** until it gets the correct **answer**. The concept of *loss* will be what helps to *optimize* the **guess**.

Task 1.1

(1) Come up with an example of where Machine Learning would be more useful than Traditional Programming. (2) Explain why using Traditional Programming would not be optimal.

An example where Machine Learning would be more useful than Traditional Programming could be when creating a large language model. Using traditional programming would not be optimal since too many rules for the english language to program using traditional methods. However, training a model on a lot of preapproved examples of the english (or any) language could be used to create a model that can output logical and readable text.

To illustrate, the below diagram shows the general machine learning process.

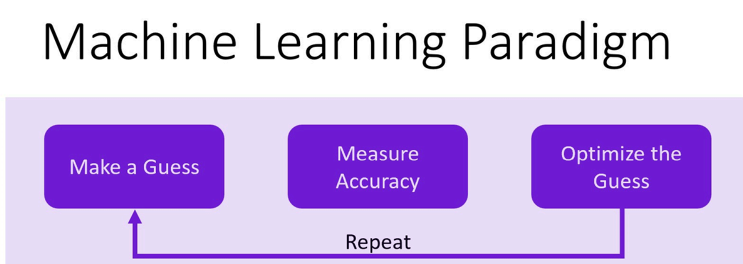


Figure 2 ML paradigm process [1]

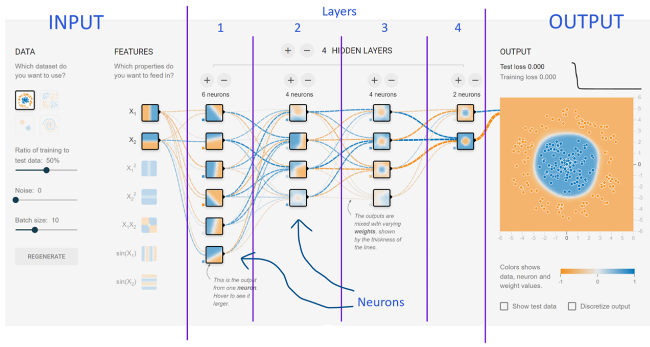
Task 1.2: (1) With one paragraph (3-5 sentences), explain machine learning very simply. Hint, using the human brain as an analogy makes sense here.

Machine learning is used when a large amount of data that follows patterns needs to be analyzed and interacted with, but the patterns are too complex or numerous for traditional programming. This large amount of data is fed into a statistical model to train it and create an algorithm to predict outcomes or outputs. The model is refined based on taking a statistical guess, comparing this guess to a true and verified value and then adjusting the models parameters for better accuracy. After training on a sufficiently large dataset the model can react with new inputs that it has not been trained on before and give accurate outputs.

Some main aspects of ML

- Neurons and Layers

- When designing a neural network, there are decisions to make on the amount of layers and the amount of neurons. Reference the picture below to see neurons & layers and how they are portrayed in a visualized ML model. If you want to play with the website used in this screenshot, go to ' https://playground.tensorflow.org/ '.



- Model
  - The product of training and data sets. When "machine learning" happens, it is usually the model making inferences on data given to it. See the figure below.

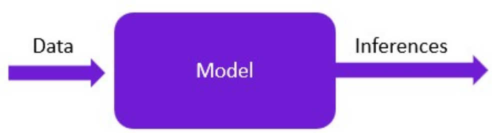
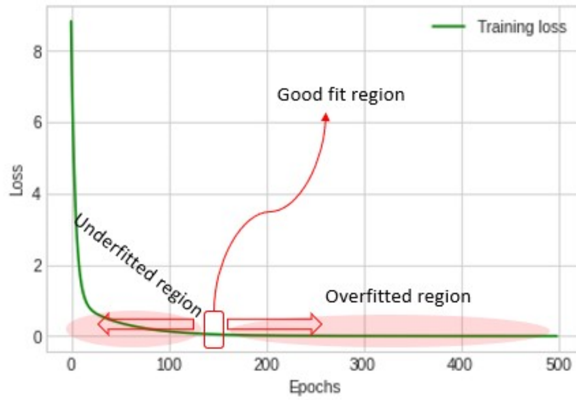


Figure 3 A "model"[1]

- Training
  - The process of determining the best "fit" for the model. After a bunch of iterations, a model will be trained for making good inferences. Seeing figure 2 above, that "Repeat" process is the training process.
- "Fit"ness
  - As an ML Engineer, you can either underfit or overfit a model. If a model is underfit, it will not make good inferences on the data used to train the model and will not make good inferences on the real-world/unknown data. An overfit model will be super good at the training data, but when in the real-world, it will not be good. Take a look at the figure below.



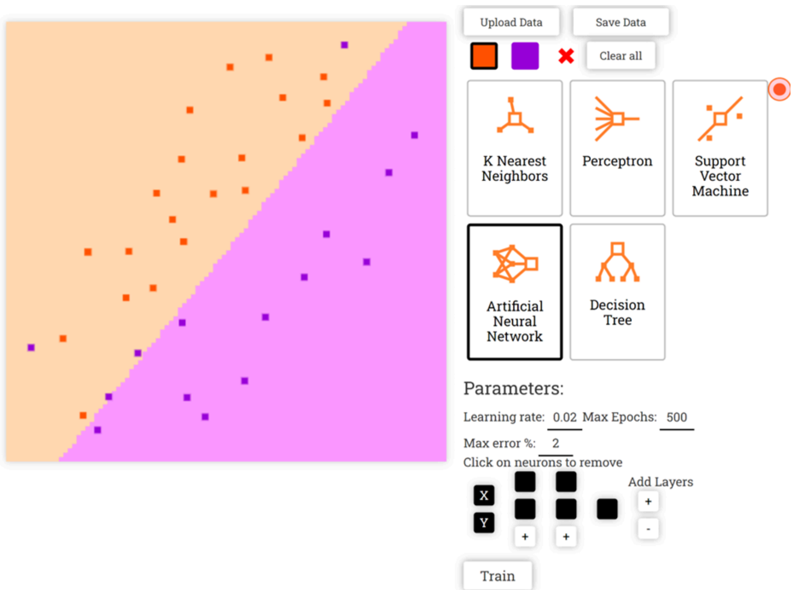
- Epochs
  - The amount of times the process in Figure 2 is repeated. When training a model, the algorithm will run through multiple iterations to minimize the loss as much as possible.
- Visualizing
  - Visualizing helps the ML Enginineer (you) determine if you have appropiately tuned your model (the parameters) according to your data set. Typically, neural networks are seen as "black boxes" -- it is hard for humans to make reliable inferences based on the machine learning code, layer decisions, neuron decisions, etc.

Exploring how neurons, layers, and algorithms change the efficiency of our chosen ML model

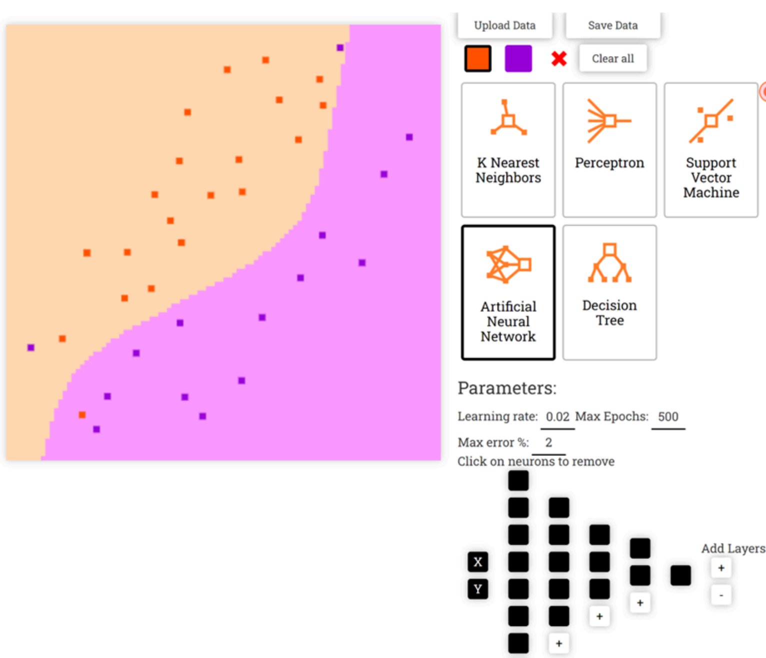
We will use the website ' https://ml-playground.com/ ' to form our intuitions and gain a better understanding of how the different neurons, layers, and machine learning algorithms effect a model.

This website allows you to click Orange and Purple dots to form data points, then choose an algoritm to train your model. Once selecting an algorithm, you'll see there are choices to let you tune your parameters. This algorithm, with parameters, is how your model will be trained. This process is visualized and simplified to help you build intuition.

The screenshot below shows a fit model. By looking at the data points, it is safe to say the algorithm should classify half orange and half purple, although there are data points, or anomalies, on either side of the opposite color. You can see in the bottom right that there are 2 layers with 2 neurons each. These parameters were chosen after trial-and-error to find the best one.



The screenshot below shows an overfit model. You see, the data is roughly split 50/50 by a diagonal line. However, the colored regions show where the model would predict a data point to be. And, in this case it would not infer correctly. It is too fit for the training data, and if we introduced new data (real-world / unknown data), it would not draw accurate inferences.



There is not an example provided for the underfit model, that is your Task.

Task 1.3

Draw data points wherever you see fit. Try to use a pattern of orange/purple that is not like the data in the screenshots we have provided.

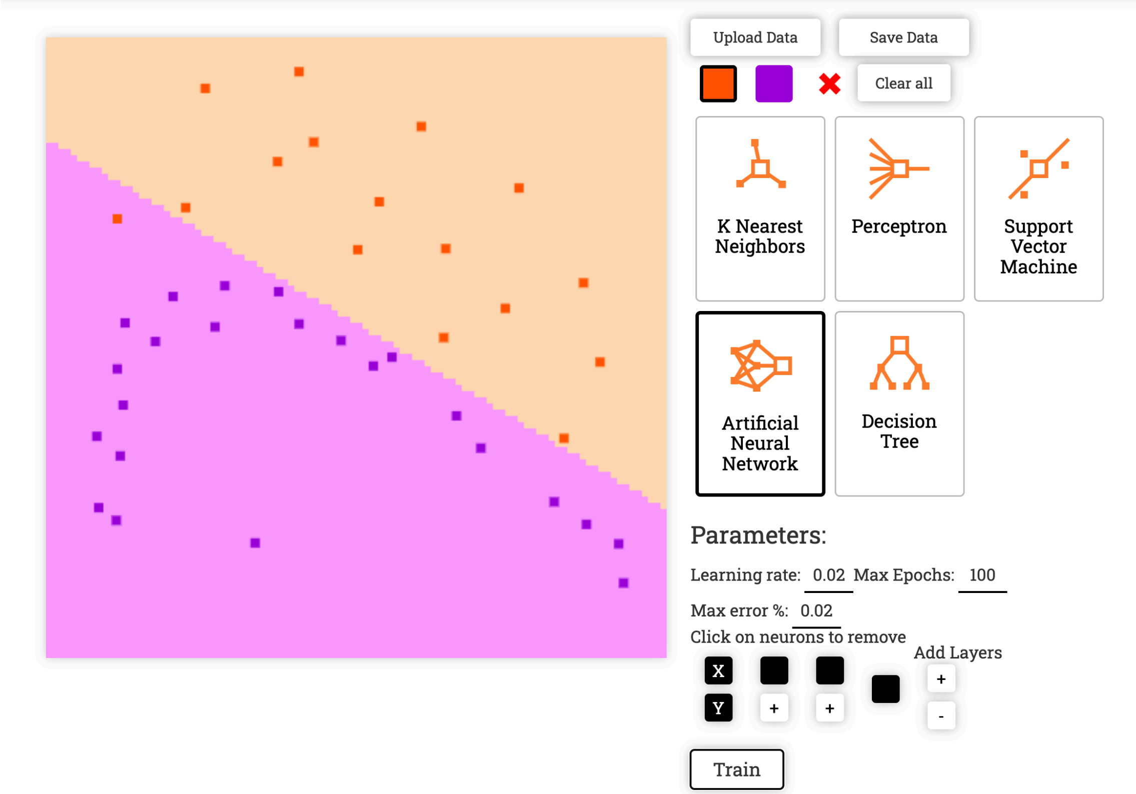
Once you have a pattern of orange/purple dots, select the **Artificial Neural Network** and mess with the parameters below. So, change the amount of neurons and layers and click "Train" to see if you think the colored regions are suitable for the data you put in. Feel free to read the information the website provides below the graph, and mess with learning rate, epochs, and error as well.

Provide a screenshot of your data with **(1)** underfit regions, **(2)** overfit regions, and **(3)** fit regions.

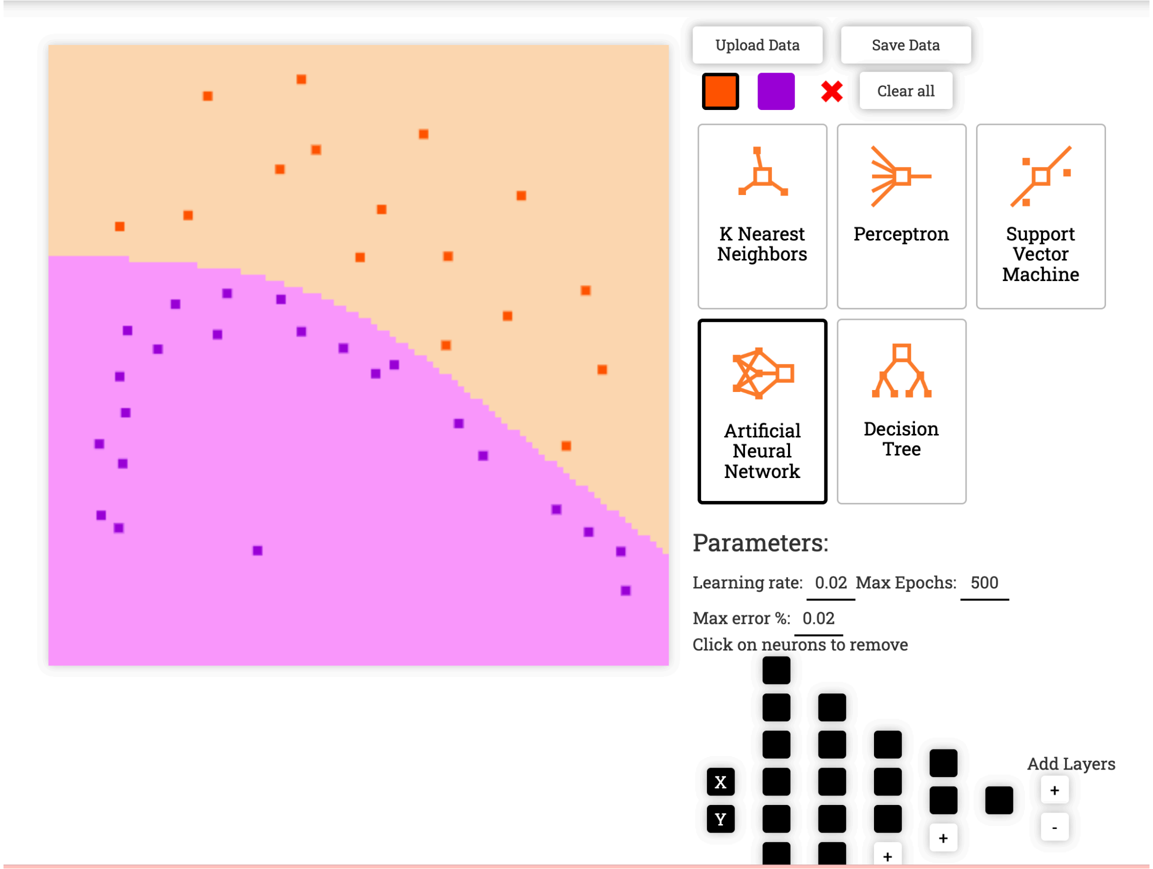
(4) Explain why the choices you made impacted your model to be fit or not fit.

Double click here, and the text form will open for edition. Paste the screenshots below each line:

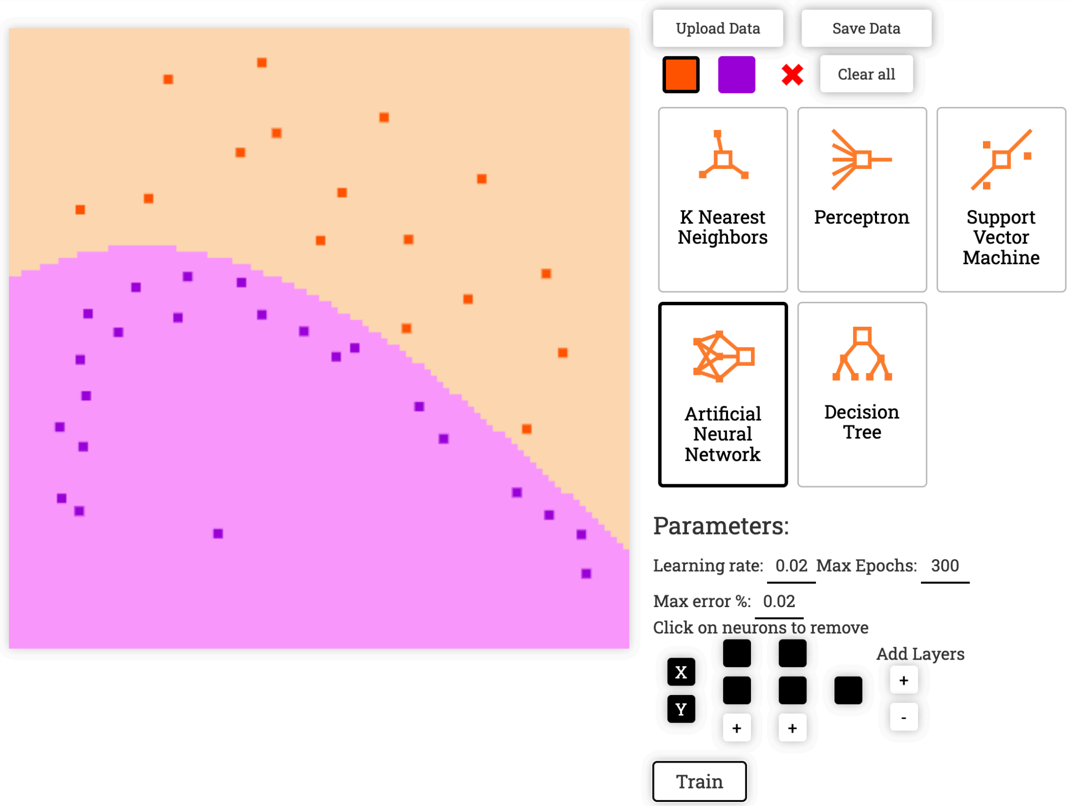
Task 1.3 (1) Underfit region:



Task 1.3 (2) Overfit region:



Task 1.3 (3)Fit region:



Task 1.4

Looking at your visual graphs, what evidence do you have to support your belief whether the model is fit/underfit/overfit? Why does the model exhibit those fit-ness characteristics?

For the underfit model, the evidence that it is underfit is that the purple and orange divided regions do not follow the shape of the data very well. It exhibits this fit-ness characteristic because it does not have enough training epochs and does not have enough neurons in each layer.

For the overfit model, the evidence that it is overfit is that the purple and orange regions are divided **too** well in that the model does not actually capture the initial slope of the purple data since it is too evenly splitting the orange and purple datasets. It exhibits these fit-ness characteristics because it has too many neurons and too many epochs.

For the fit model, the evidence that it is fit is that the model correctly captures the slope of the purple dataset. It has an appropriate number of training epochs and neurons in each layer.

Interpreting Air-tight Vacuum and Air-leak Vacuum Data

Proceed to the next Colab Notebook to work in building a model on anomaly detection in a vacuum pump example in Prelab 8.2.

Please continue to Prelab 8.2 here.